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ABERDARE TECHNICAL COLLEGE
INFORMATION COMMUNICATION TECHNOLOGY
UNIT: PERFORM PROGRAMMING PRINCIPLES

UNIT CODE: 0613 451 05A

CAT 1

MODULE 4, TERM 4

TOTAL: 50 MARKS

TIME 1 HOURS 30 MIN

NAME KARAU MARCUS.....CDACC REG.....

DATE 16/09/2020.....SIGN [Signature].....

Instructions

- ✓ This paper consists of **TEN (10)** questions and carries a total of **50 marks**.
- ✓ Answer **ALL** questions in the spaces provided.
- ✓ The time allowed for this paper is **1 hour**.
- ✓ Write your **Name, Admission/Registration Number** clearly on this sheet before you begin.
- ✓ Marks allocated to each question are shown in brackets at the end of the question or sub-question.
- ✓ Read each question carefully before answering. Where an example is required, ensure it is relevant and correctly stated.

QUESTIONS

1. a) Define the term "programming language." (2 marks)

b) Programming languages can be grouped into different categories. Briefly describe **three** of the following categories, giving one example language for each: (5 marks)

- Procedural languages
- Object-oriented languages
- Functional languages

2. Distinguish between **low-level** and **high-level** programming languages, giving one example of each. (4 marks)

3. State **four** criteria that should be considered when selecting a programming language to meet a given user's requirements (4 marks)

4. A hospital wants to develop a system for managing patient records, appointments, and billing. As the systems developer, identify and explain **five** criteria you would use to select an appropriate programming language for this system. (10 marks)

(Relate each criterion directly to the scenario.)

5. a) Define the term "programming paradigm." (2 marks)

1. A Programming language is a set of ^{instructions} that is used to ~~write~~ Program on Computer to understand and execute.
2. Procedural languages: Programs are written as a sequence of step by step procedures or functions.
- Object oriented language: Programs are organized around objects that contain both data and methods that operate on that data.
- Functional Languages: - Programs are built by applying & composing pure functions with little or no changeable state.

On Examples?

2. Low level programming languages are close to computer machine languages.
- High level programming languages are close to human languages.

3. Suitability for the problem.

Ease of learning and use by development team.

Performance requirements.

Portability requirements.

4. Data security and privacy → The target system must support strong security features and data encryptions.

Database Support: → System should have good frameworks for reliable database access, storage and retrieval.

Ease of maintenance and scalability → The system should have maintainable code that can be updated as new features are added.

Availability of skilled developers → There should be developers familiar with the language for easy update and maintain.

Relate each criterion to the scenario

5. Programming Paradigm: is a formal approach for writing computer programs.

Paradigm

6. Procedural → Solve problems step by step sequence.

Object oriented Paradigm → Solve by modeling it as interacting objects that contain both data and the operations that work on that data.

Functional Paradigm -> Solves Problems by Composing Pure functions that receive inputs and produce output without changing external state. 8)

2. F. ~~Know~~ ^{Know} ~~how~~ ^{how} ~~it~~ ^{it} ~~solves~~ ^{solves} ~~problems~~ ^{problems}. Then proceed to how it solves problems.

7. Functional Programming: is programming that which computation is treated as the evaluation of mathematical functions and avoid changing state.

b. Use of Pure functions. ✓ 8)

Emphasis on immutability. ✓ 8)

8. Procedural Programming: is a style that organizes codes into procedures that operate on data by following a sequence of steps. 13)

3. Simple and easy to understand. ✓

Efficient for problems that can be broken down into clear sequential steps. 8)

d. Object-oriented programming is a style that organizes software design around objects, which are instances of classes that encapsulate both data and methods that operate on the data. ✓ 8) 8)

Encapsulation: bundling data & methods together

10. Inheritance -> Allowing a new class to reuse and extend the properties and behavior of an existing class.

10. Distinguish.

Imperative Programming - this tells the computer how to do something by giving step by step commands. 8)

Declarative Programming -> Tells the computer what is wanted without any step by step specification for the result. 8)

Give Example in each